

DTSA 5843

Effective Communication Capstone Project

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (45.3 / 100)	35.2 / 100
SPECIALIZATION Effective Communication		

35

CURRENT AVG SCORE
across 6 job roles

71

POTENTIAL SCORE
with recommended additions

+56

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

In this course students will create a portfolio of work that demonstrates their mastery of writing, design, and presentation skills. The portfolio includes three elements—a memo, a slide deck, and deliver presentation—integrated around a single topic. The capstone allows learners to engage meaningfully in their world by choosing a project relevant to their job. Effective Communication: Writing, Design, and Presentation is a prerequisite for this course.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	45.3	77.6	+32.3	HIGH
Applied Scientist	44.5	100.0	+55.5	HIGH
AI Engineer	37.3	70.9	+33.6	HIGH
Data Scientist	32.9	69.0	+36.1	HIGH
ML Engineer	28.6	58.6	+30.1	HIGH
MLOps Engineer	22.9	52.7	+29.8	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

HIGH

Add System Design coverage — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2

LOW

Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

45.3 → 77.6 (+32.3 pts)

data sets learning analytics tools exploratory business mobile

Applied Scientist

44.5 → 100.0 (+55.5 pts)

lab discovery scientific research literature interdisciplinary models technical

AI Engineer

37.3 → 70.9 (+33.6 pts)

perception following data models learning technical systems teams

Data Scientist

32.9 → 69.0 (+36.1 pts)

data sets analytics social

ML Engineer

28.6 → 58.6 (+30.1 pts)

media discovery data learning machine systems models new

MLOps Engineer

22.9 → 52.7 (+29.8 pts)

data text systems learning machine media tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.